

LUAD Top-10 identity and evidence matrix

Review date: 2026-09-25. This is a research triage artifact, not a treatment recommendation. The screen used one A549 perturbation condition (10 μ M, 24 h). Replaying the published EH3226 meta-data filter recovered one exact GSE92742 sig_id and pert_id for every Top-10 row; the crosswalk is configs/luad_top10_signature_ids.csv. Cross-source chemical conflicts with the dated [Broad Drug Repurposing Hub](#) remain where noted.

Candidate-level audit

Rank 1 · hydrocortisone

- **Identity status:** Exact source signature BRD-A75172220; Hub cross-source stereochemistry differs
- **Name-level targets / mechanism:** ANXA1, NOS2, NR3C1, NR3C2; glucocorticoid-receptor transcription and annexin/eicosanoid signaling
- **LUAD/NSCLC evidence found:** Hydrocortisone shared GR-dependent A549 growth arrest with dexamethasone in vitro ([PMID 1533045](#)); an independent network study proposed hydrocortisone computationally for NSCLC ([PMID 35676421](#))
- **Evidence against or uncertainty:** GR activation has also been linked to chemotherapy resistance and stemness ([PMID 41344105](#)); no candidate-specific LUAD animal or clinical efficacy study was found
- **Decision:** Insufficient; source ID resolved, cross-source chemistry and efficacy unresolved

Rank 2 · beclomethasone-dipropionate

- **Identity status:** Exact source signature BRD-K97810537; exact Hub InChIKey match
- **Name-level targets / mechanism:** CYP3A5, GPR97, NR3C1; GR-regulated transcription and xenobiotic metabolism
- **LUAD/NSCLC evidence found:** Beclomethasone dipropionate induced CYP3A5 through GR in A549 cells ([PMID 12538830](#)); its active monopropionate metabolite also induced CYP3A5 in A549 ([PMID 24555085](#))
- **Evidence against or uncertainty:** These are pharmacology observations, not antitumor efficacy. No LUAD animal or clinical efficacy study was found; class-level chemoresistance concerns remain
- **Decision:** Insufficient; verified identity and A549 target engagement, no efficacy evidence

Rank 3 · clocortolone-pivalate

- **Identity status:** Exact source signature BRD-K38003476; exact Hub InChIKey match
- **Name-level targets / mechanism:** NR3C1, PLA2G1B; glucocorticoid-receptor agonism and phospholipase/eicosanoid context
- **LUAD/NSCLC evidence found:** No drug-specific LUAD/NSCLC efficacy study was found in the bounded review; only class-level GR evidence applies

- **Evidence against or uncertainty:** No compound-specific validation; topical dermatologic use does not establish tumor exposure or a feasible oncology dose
- **Decision:** Insufficient

Rank 4 · mometasone

- **Identity status:** Exact source signature BRD-K60640630; GEO/Hub structure mismatch remains
- **Name-level targets / mechanism:** NR3C1; glucocorticoid-receptor agonism
- **LUAD/NSCLC evidence found:** No drug-specific LUAD efficacy evidence was found; inhaled-corticosteroid observational literature is only class-level prevention context
- **Evidence against or uncertainty:** Cross-source structures disagree, so name-level literature may describe another formulation. No treatment-efficacy evidence
- **Decision:** Insufficient; cross-source identity mismatch is a hard stop

Rank 5 · flumetasone

- **Identity status:** Exact source signature BRD-A01346607; synonym mapping remains ambiguous
- **Name-level targets / mechanism:** Likely flumethasone-class NR3C1 agonism; some Hub synonyms also annotate PLA2G1B or SERPINA6
- **LUAD/NSCLC evidence found:** No exact-name LUAD/NSCLC efficacy study was found
- **Evidence against or uncertainty:** flumetasone may refer to flumethasone or an ester formulation; target transfer across synonyms is not chemically verified
- **Decision:** Insufficient; synonym resolution required

Rank 6 · BRD-K84203638

- **Identity status:** Exact source signature and GEO structure recovered; no Hub sample
- **Name-level targets / mechanism:** Unknown
- **LUAD/NSCLC evidence found:** No linked LUAD efficacy evidence was found
- **Evidence against or uncertainty:** The recovered structure and PubChem field permit future reconciliation, but no verified target or oncology study currently supports interpretation
- **Decision:** Insufficient; identity improved, biology unresolved

Rank 7 · hydrocortisone-hemisuccinate

- **Identity status:** Exact source signature BRD-A07000685; Hub cross-source stereochemistry differs
- **Name-level targets / mechanism:** Parent-drug inference suggests NR3C1/NR3C2 and ANXA1 signaling; the Hub name row has no target annotation
- **LUAD/NSCLC evidence found:** Hydrocortisone class evidence provides mechanistic context only
- **Evidence against or uncertainty:** Salt/ester and cross-source stereochemistry remain unresolved; hydrocortisone evidence cannot be promoted to this exact material
- **Decision:** Insufficient

Rank 8 · fluorometholone

- **Identity status:** Exact source signature BRD-A13133631; no Hub sample
- **Name-level targets / mechanism:** NR3C1; glucocorticoid-receptor agonism

- **LUAD/NSCLC evidence found:** No drug-specific LUAD/NSCLC efficacy evidence was found
- **Evidence against or uncertainty:** No Hub sample-level match and no oncology validation; ophthalmic anti-inflammatory use is not evidence of tumor exposure
- **Decision:** Insufficient

Rank 9 · fluticasone

- **Identity status:** Exact source signature BRD-K14791739; formulation mapping across sources remains ambiguous
- **Name-level targets / mechanism:** Formulation-dependent annotations include NR3C1, NR3C2, PGR and PLA2G4A
- **LUAD/NSCLC evidence found:** Observational studies of inhaled corticosteroids report possible lower lung-cancer incidence, including fluticasone-equivalent dose analyses ([PMID 35843928](#))
- **Evidence against or uncertainty:** Prevention associations are low/very-low certainty and do not establish treatment of existing LUAD. Systemic GR effects may conflict with immunotherapy
- **Decision:** Insufficient; prevention context only

Rank 10 · diflorasone

- **Identity status:** Exact source signature BRD-K17674993; no Hub sample and name variants remain
- **Name-level targets / mechanism:** NR3C1 for diflorasone-diacetate annotations
- **LUAD/NSCLC evidence found:** No drug-specific LUAD/NSCLC efficacy study was found
- **Evidence against or uncertainty:** Cross-source formulation mapping remains unresolved, and no sample-level Hub match exists
- **Decision:** Insufficient

Shared pathway evidence and contradiction

Nine of ten names are known or likely glucocorticoids; BRD-K84203638 is unresolved. This explains a coherent transcriptional class signal, but it also means the Top-10 is not ten independent mechanistic hypotheses.

- Supporting preclinical context: dexamethasone suppressed A549 and Calu-1 proliferation through ERK/MAPK and cell-cycle regulation ([PMID 12204894](#)). A biomarker-selected study found dexamethasone responses in STK11/LKB1-mutant NSCLC models ([PMID 37035141](#)). These findings are class-level and biomarker-dependent; they do not validate the nine screened compounds.
- Contradicting preclinical context: dexamethasone reduced pemetrexed cytotoxicity in GR-responsive nonsquamous NSCLC models ([PMID 24736075](#)). A later study linked GR/ β -catenin activation to lung-cancer stemness, chemoresistance, worse survival signals, and reduced paclitaxel benefit ([PMID 41344105](#)).
- Clinical-context uncertainty: observational inhaled-corticosteroid studies sometimes report lower lung-cancer incidence, but a systematic review found no significant effect in the included randomized trials ([PMID 27761973](#)); a later meta-analysis rated the evidence low to very

low ([PMID 35843928](#)). This is prevention evidence in COPD/asthma populations, not LUAD treatment evidence.

- Immunotherapy context: retrospective NSCLC studies associate baseline or early systemic steroids with worse checkpoint-inhibitor outcomes, although indication-related confounding is material and some analyses differ ([PMID 33007977](#), [PMID 33431285](#), [PMID 36142204](#)).

Bottom line

The reconstruction resolves the exact source signature and perturbagen IDs, but it does not produce a clinically credible Top-10. Cross-source formulation or stereochemistry conflicts remain for several rows. Only hydrocortisone and beclomethasone dipropionate have candidate-specific A549/NSCLC context, and neither has candidate-specific LUAD efficacy evidence. The appropriate output remains a hypothesis list with `insufficient_evidence`. The prespecified dose, subtype, mechanism and combination plan is in `docs/luad_experimental_validation_protocol.md`.

The search was deliberately bounded to exact names, likely formulation synonyms, A549/LUAD/NSCLC terms, and shared glucocorticoid evidence. “No study found” means none was identified in this bounded review, not proof that no literature exists.